

Bayesian Branches

Where do language trees come from?

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Bayes Bro #3 (derogatory)

28/09/2026 · SMG Expertise Session



What a tree is (and is not...)

What a tree is

A (genealogical) tree, a.k.a. **phylogeny**, is a **hypothesis** about descent. It's a **model** of the past, involving:

- **Nodes**

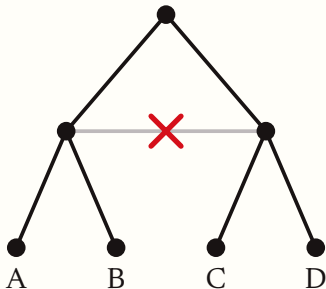
- Tips: observable languages
- Internal nodes: inferred ancestors
- Root: the common ancestor to all

- **Branches**

- Lines of descent from ancestor to descendant
- length = time, amount of change, or nothing

- **One rule:** 1 parent per node (except root)

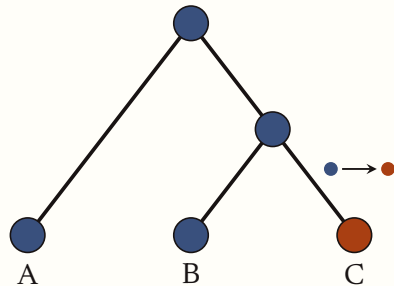
- Only vertical descent allowed
- (>1 parent = network. More on this later.)



What a tree is not

Phylogenetic tree $\neq$ **similarity** tree (a.k.a. dendrogram)

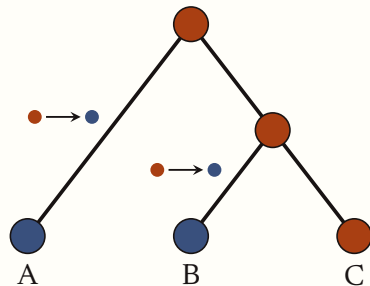
- Languages are grouped by **shared history**, not overall resemblance
- Resemblance can be misleading in two ways:
 - **Shared retentions**: old traits some lineages kept and others lost



What a tree is not

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- Languages are grouped by **shared history**, not overall resemblance
- Resemblance can be misleading in two ways:
 - **Shared retentions**: old traits some lineages kept and others lost
 - **Convergence**: similar traits arising independently



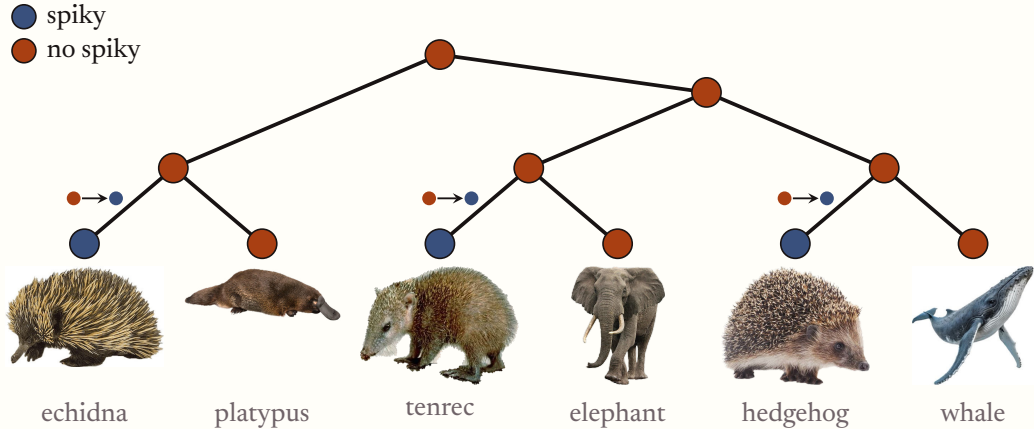
What a tree is not

- Only **shared innovations** are evidence of common descent

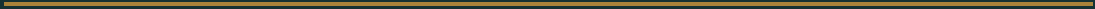
Sound familiar?

Subgrouping via the comparative method is also based on shared innovations, not retentions.

Convergence in the wild: spikybois



Feeding the BEAST



Data

- Most language trees are built on **lexical data**
 - **Cognate sets**, typically from 100–200-item Swadesh-style basic vocab lists
 - Related wordforms are identified, then binarised into **characters**.
- A language switching to a new word for a given meaning is a **lexical innovation**.
- In principle, anything **inherited with modification** could work (sound changes, morphology, syntax).
- In **biology**...
 - Today: mostly genomic sequences.
 - Historically (and still today for fossils, like dinosaurs): morphology.

Cognates to characters

Language	<i>cow</i>	Class
English	cow	A
German	Kuh	A
Irish	bó	A
French	vache	B
Spanish	vaca	B
Russian	korova	C
Polish	krowa	C



Language	<i>cow</i>		
	A	B	C
English	1	0	0
German	1	0	0
Irish	1	0	0
French	0	1	0
Spanish	0	1	0
Russian	0	0	1
Polish	0	0	1

- A PIE *g^wóws
- B Latin *vacca*
- C Balto-Slavic *karwā

A meme, as a treat (you've earned it)

POV: you told your buddy this was a good place to pasture your *g^wōwns, but now you see a Männerbund menacingly riding towards you.



Where linguistics happens

Cognate coding is usually “the hard part”, requiring manual effort and actual **linguistic judgement**.

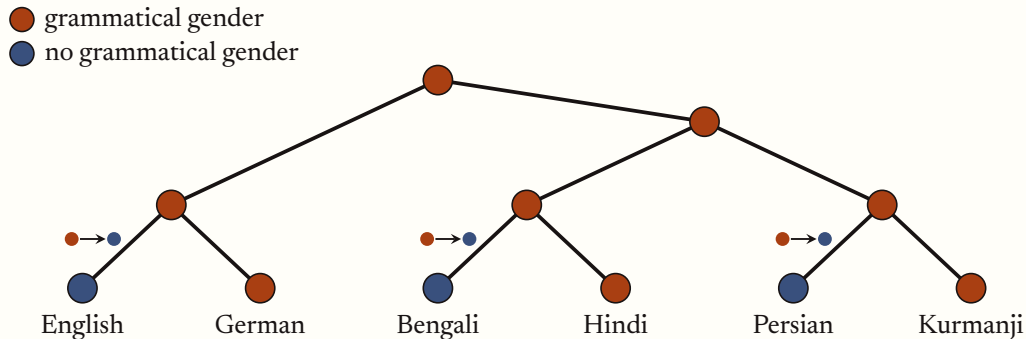
- Cognacy requires **regular sound correspondences**, not just similarity. (Circular? Maybe... Let's talk later.)
- **Borrowings** must be filtered out.
- **Semantics = pain**. (Ensuring meanings are comparable, dealing with (near-)synonyms, semantic shift).
- **Partial cognacy** (compounds, shared roots with different derivational material).
- Documentation gaps, **missing data**.
- **Automated cognate detection** (e.g. Mattis List's *LexStat*) scales up coding, but still wants expert checking.

Inside the black box

Parsimony

- Choose the tree that requires the **fewest changes** to explain the data
- Appealing:
 - Intuitive; Occam's razor
 - Computationally super cheap
 - Dominant method for decades; still useful for quick trees
- But it treats every change as equally unlikely, regardless of how often that kind of change happens or how long a lineage has had to change
- Where there is **convergence**, the fewest-changes tree can be the wrong tree

Parsimony and grammatical gender



The fix

- Don't count changes – **model** how change happens
- Each character is **gained** ($0 \rightarrow 1$) and **lost** ($1 \rightarrow 0$) along branches, at some rate
- This gives us a **likelihood**:

$$P(\text{data} \mid \text{tree}, \theta)$$

- How probable is our cognate matrix, if this tree (with these branch lengths) were true?
- θ = the evolutionary model: **transition rates** (and other things)
- Long branch, fast rate $\Rightarrow$ more convergence **expected**, modelled rather than just penalised

Maximum likelihood

- Goal: the **single** tree (and θ) that maximises $P(\text{data} \mid \text{tree}, \theta)$
- Too many trees to check them all, so **climb**:
 - Start somewhere, try rearranging branches, keep any improvement
 - Repeat until nothing improves; restart elsewhere to avoid getting stuck on a lower peak
- Output: one best tree, with θ fixed at its best values
- Uncertainty via **bootstrapping**:
 - Resample the data columns, rerun, repeat (e.g. 1,000 times)
 - “Bootstrap support 87%” = clade found in 87% of reruns
- Fast, widely used in biology (IQ-TREE, RAxML)
- But: no priors, so no natural home for **dates** or outside evidence

Tree inference for Bayes bros

Bayes' theorem, for trees

The diagram illustrates Bayes' theorem for trees. The equation is $P(\text{tree}, \theta \mid \text{data}) = \frac{P(\text{data} \mid \text{tree}, \theta) P(\text{tree}, \theta)}{P(\text{data})}$. The components are labeled as follows:

- posterior: what we want**: Points to the numerator $P(\text{tree}, \theta \mid \text{data})$ with a black arrow.
- likelihood (seen it!)**: Points to the term $P(\text{data} \mid \text{tree}, \theta)$ in the numerator with a blue arrow.
- prior**: Points to the term $P(\text{tree}, \theta)$ in the numerator with a red arrow.
- marginal likelihood**: Points to the denominator $P(\text{data})$ with a grey arrow.

The terms in the equation are enclosed in rounded rectangles: $P(\text{tree}, \theta \mid \text{data})$ is black, $P(\text{data} \mid \text{tree}, \theta)$ is blue, $P(\text{tree}, \theta)$ is red, and $P(\text{data})$ is grey.

So... what goes in the **prior**?

Priors

- **Informative:** things we already know
 - **Clade constraints:** subgroups established by the comparative method
 - **Calibrations:** dated ancestors or splits (archaeology, historical records)
 - **Ancient languages** as dated tips (e.g. Hittite, Latin, Sanskrit)
- **Structural:** what trees and change are generally like
 - **Tree prior:** how lineages split (and die out) over time
 - **Clock:** rates of change, and if/how these rates vary across branches
- **Uninformative:** where we know little
 - Transition rates, rate variation across characters: wide ranges, let the data speak

One problem...

$P(\text{data}) = \text{likelihood} \times \text{prior}$, summed over **every possible tree** (and every θ)

How many possible (rooted, bifurcating) trees are there?

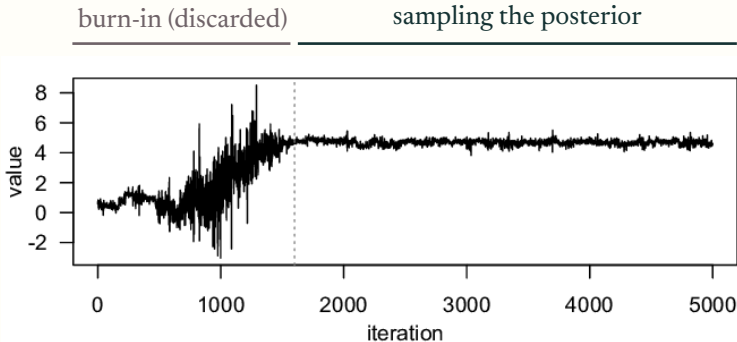
Languages	Possible trees
3	3
5	105
10	34,459,425
20	8×10^{21} (more than grains of sand on Earth)
52	3×10^{80} (atoms in the observable universe)
100	3×10^{184} (a typical Indo-European study)

... and that's **before** branch lengths and rates.

The solution: MCMC

- We can't **compute** the posterior, but we can **sample** from it
- Markov chain Monte Carlo: a random walk through tree space
 - Propose a small change: swap some branches, nudge a branch length or rate
 - Better? Move. Worse? Move **sometimes**: half as probable, move half the time
 - Only ever needs the ratio of two posteriors, so $P(\text{data})$ cancels out
- Repeat millions of times: how often a clade turns up $\approx$ its **posterior probability**

Witchcraft



- More trees than atoms in the universe, yet it finds the right neighbourhood after visiting a vanishingly small fraction of them
- Trust, but verify: run several chains and check they agree

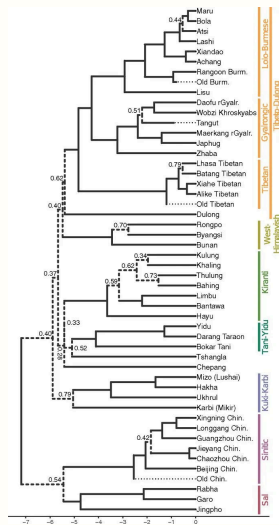
See the forest for the trees

What comes out

- A **posterior sample** of trees: a “forest”
 - Saved every n th step of the chain, after burn-in (e.g. 1,000–10,000 trees)
 - Every tree has its own topology, branch lengths and θ
- A posterior distribution for **every parameter**
 - Rates, clock, ...
 - **Root age** and **clade ages**: a most probable range, not a single date
 - Reported as e.g. “95% HPD”: the narrowest range holding 95% of the posterior
- **Clade support**: the proportion of sampled trees containing a clade

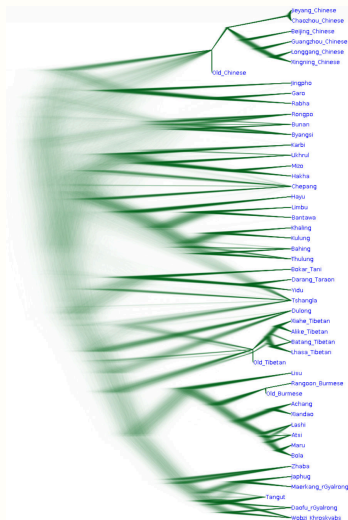
Summary trees

- A single tree in a Bayesian paper is a **summary** of the forest, not *the* answer
- Several ways to summarise, e.g.:
 - **Maximum clade credibility (MCC)**: score each clade by how often it appears in the forest; pick the sampled tree with the highest product of its clade scores
 - **Majority-rule consensus**: keep clades found in $>50\%$ of trees
- Node labels: **clade support** (0.40 = in 40% of trees in the forest)
- Be wary of a single tree with **no support values**



DensiTree

- Draw **every** sampled tree on top of each other, semi-transparent
- Dark and sharp: the trees agree
- Faint and fuzzy: **uncertainty**
- Not a network! Criss-crossing = sampled trees **disagreeing**, not language contact



Embracing uncertainty

- Modern comparative methods put relatedness **directly into the statistical model**
 - No need for balanced sampling: keep **all** the data
 - Even an imperfect tree beats no tree (Macklin-Cordes & Round 2022)
- But which tree? A single summary tree treats one guess as certain
- Better: repeat the analysis over a **sample of trees** from the posterior, and combine
- Uncertainty about history flows through into uncertainty about the result

Case study: The Great Indo-European wars

Anatolian vs Steppe hypotheses

- **Steppe** (Kurgan) hypothesis
 - Pontic–Caspian steppe; pastoralists with horses and wheels
 - Dispersal c. 6500–5500 BP
 - The historical linguists' favourite (wheel vocabulary, contact with Uralic, ...)
- **Anatolian** (farming) hypothesis
 - Neolithic Anatolia; spreads with agriculture (Renfrew 1987)
 - Dispersal c. 9500–8000 BP
- 2003: a Bayesian tree backs **Anatolia** (Gray & Atkinson 2003). Uproar ensues.



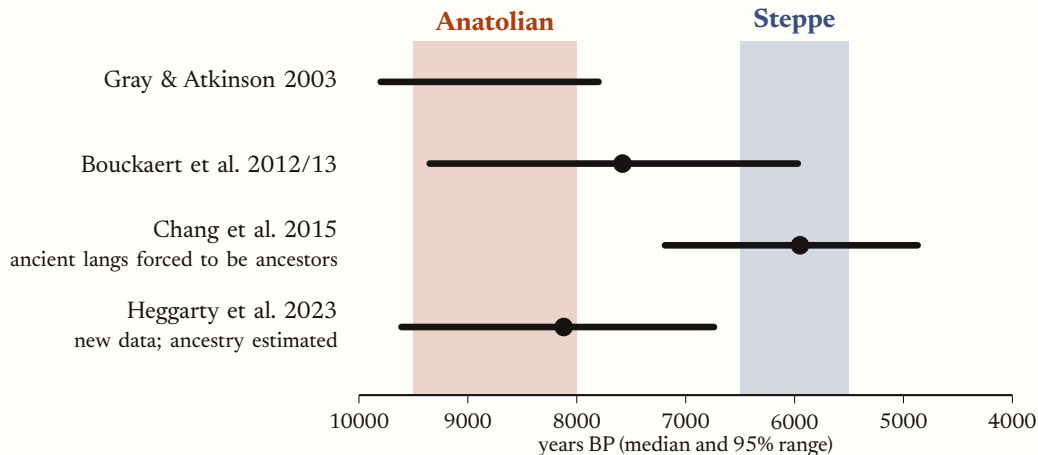
Bouckaert et al. (2012)

Meanwhile, on the steppe



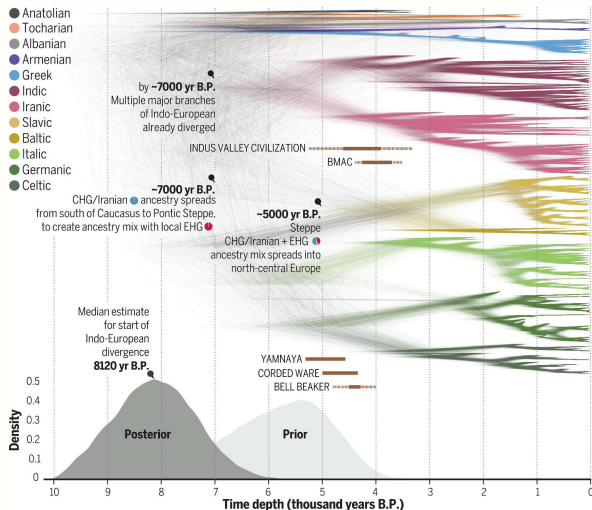
A dating question

The two hypotheses predict different **root ages**. So: estimate the root age, see which window it lands in.



Everyone's a winner?

- Heggarty et al. (2023): recoded dataset, 109 modern + 52 ancient languages
- Ancestry **tested, not imposed**: few ancient languages are direct ancestors
- Root c. 8120 BP: too old for the Steppe
- A **hybrid**: origin south of the Caucasus; one branch north to the steppe, then into Europe c. 5000 BP
- Note the prior *vs* posterior for root age!



Not over yet

- Critics: dataset and method (Kassian & Starostin 2025); dates clash with archaeologically datable wheel/wagon vocabulary (Anthony & Ringe 2015)
- Ancient DNA (Lazaridis et al. 2025): Indo-Anatolian ancestors in the Caucasus–Lower Volga c. 6400–6000 BP; Yamnaya behind the rest
- **Where:** rough agreement on the Caucasus region (though which side is disputed)
- **When:** the tree's root is ~2000 years older than the ancient-DNA estimate
- Absolute dates are the most fragile output of a tree

What about contact?

Linguistic exceptionalism

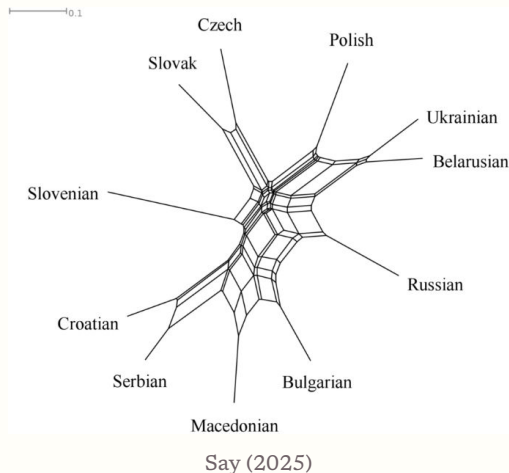
- “Trees can’t work for languages: languages borrow!”
- Biology is messy too:
 - **Hybridisation**: common in plants; non-Africans carry ~2% Neanderthal DNA
 - **Horizontal gene transfer**: rampant in bacteria; viruses recombine
 - **Incomplete lineage sorting**: different genes, different trees (cf. different words, different histories)
- But languages **are** different:
 - More horizontal transmission, faster, often socially driven
 - Smaller, hand-coded, messier datasets
- So the question isn’t “is it a tree?” but “how tree-like, and does it matter for my question?”

Trees vs waves

- **Wave model** (Schmidt 1872): innovations spread across a dialect continuum, giving overlapping isoglosses rather than nested subgroups
 - Fair at some scales: in-situ diversification of dialect networks (François 2014, Kalyan & François 2018)
 - Different process, not a refutation: many families show clear nesting, and basic vocabulary is mostly inherited

Networks and tree-likeness

- **Networks** (e.g. NeighborNet) visualise conflicting signal
 - Exploratory: a box is conflict in the data, not a contact event
- **Measure** tree-likeness: delta scores, Q-residuals (Gray, Bryant & Greenhill 2010)



Current approaches

- **Filter** known borrowings; rate-variation models soak up some noise
- **Robustness**: in simulations, realistic borrowing leaves topology largely intact but biases dates younger (Greenhill, Currie & Gray 2009)
- **Model contact explicitly**: contacTrees (Neureiter et al. 2022), a tree plus borrowing edges in BEAST 2; recovers known IE contacts and loanwords
- Why so hard?
 - Network space dwarfs tree space (already beyond atoms in the universe)
 - Each word can take its own path through the network: likelihood grows exponentially with contact edges
 - Contact, shared inheritance and parallel change can explain the same data

Recap

- A tree is a **hypothesis** about descent: only shared innovations count
- Linguists shouldn't just feed the model; we should help **build** it
- Don't count changes, **model** them
- Bayesian inference = likelihood $\times$ prior; we **sample** the posterior with MCMC because we can't compute it
- The output is a **forest**, not a tree: use the uncertainty
- Trust **topology** more than **dates**

Questions & Discussion

Questions & discussion

Almost all of this rests on lexical data. What about other data types?

Why hasn't grammatical or morphological data worked, and what would it take?

Questions & discussion

You're reading a typology paper that controls for phylogeny using a sample of 1,000 dated Bayesian trees. Which parts of those trees would you trust: the topology, the relative branch lengths, the absolute dates? For their conclusions, does it matter?

Would it change your view if a different method, such as maximum likelihood or a Glottolog classification, gave a different tree?

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Thank you

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